

Multi-Currency Expense Tracker

A PROJECT REPORT

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CHAPTER 1. INTRODUCTION

1.1 Introduction to Project

The Multi-Currency Expense Tracker is a smart and user-friendly desktop application built using JavaFX. It is specifically designed to help individuals monitor and manage their day-to-day expenses across multiple currencies. In the modern era of globalization, many people encounter financial transactions in different currencies due to international travel, online purchases from global websites, studying abroad, or freelancing for foreign clients. Unfortunately, most existing expense-tracking tools are limited to handling expenses in a single currency, which creates confusion and a lack of financial clarity for users dealing with multiple currencies.

This project addresses this challenge by offering a well-organized system where users can input expenses by specifying the amount, category, currency, date, and a short description. The standout feature of the application is its ability to convert all recorded expenses into a single base currency (like INR, USD, EUR, etc.), based on the user's preference. This conversion gives users an accurate picture of their total expenditure without needing to perform any manual currency conversions or use external tools.

In addition to currency conversion, the app provides a real-time table that displays all the added expenses and a pie chart that visually categorizes spending based on user-selected expense categories (e.g., food, travel, bills, shopping). These visual insights allow users to better understand their financial habits and identify areas where they might be overspending.

The application is developed using JavaFX and includes clean UI components like ComboBoxes, TextFields, DatePickers, and PieChart. It is lightweight, responsive, and works entirely offline, ensuring privacy and fast access. It's particularly useful for international students, freelancers, travelers, and remote workers, making financial tracking easy, centralized, and visually appealing. Overall, the Multi-Currency Expense Tracker simplifies personal finance by offering a modern, currency-aware solution for real-world money management challenges.

1.2 Identification of Problem

In the context of modern finance and global interaction, individuals frequently face the challenge of managing expenses across different currencies. Whether it's an international student paying tuition fees in USD while spending locally in INR, or a freelancer receiving payments in EUR and using that income to make purchases in JPY or GBP, currency inconsistency becomes a major issue. Traditional expense trackers fail to accommodate this, as they generally allow users to log expenses in only one default currency, lacking the flexibility to record, view, and analyze multi-currency transactions in a meaningful way.

This results in financial mismanagement and confusion. Users are left either converting every transaction manually, maintaining separate logs for each currency, or worse, ignoring conversions altogether—leading to an inaccurate understanding of their actual expenses. Moreover, calculating the total amount spent in a common currency becomes time-consuming and error-prone without

proper tools. This is especially problematic for people dealing with frequent online purchases, global services, or foreign travel, where exchange rates can vary daily.

Another key limitation of existing systems is the lack of visualization and instant feedback. Many apps do not offer dynamic tools like pie charts, real-time total calculations, or category-wise analysis. Users do not get actionable insights about their spending patterns, which weakens the core purpose of financial tracking.

The need for a multi-currency, real-time, offline expense tracking application is therefore evident. The system must allow users to:

- Enter expenses in various currencies
- Automatically convert each into a base currency
- View totals and spending breakdowns
- Use intuitive visuals to enhance financial awareness

The Multi-Currency Expense Tracker aims to solve these problems. It fills the gap in existing tools by offering an efficient and clear way to handle finances in today's currency-diverse personal and professional life.

CHAPTER 2. BACKGROUND STUDY

2.1 Existing Solutions:

There are several personal finance and expense tracking tools available in the market, such as Mint, YNAB (You Need A Budget), Money Manager, Wallet by BudgetBakers, and Expensify. These applications are designed to help users monitor their spending habits, set budgets, and categorize expenses. While they do offer useful features like bank syncing, chart visualization, and budget reminders, most of them are heavily dependent on internet access and user account creation. Furthermore, they are primarily optimized for mobile platforms, and their desktop versions are either unavailable or lack full functionality.

When it comes to multi-currency support, very few applications offer built-in capabilities. Apps like Expensify and Wallet support multiple currencies but often hide these features behind premium subscriptions. Even in those cases, they do not offer real-time toggling of base currency for consolidated reporting. Additionally, the user experience can be cluttered due to excessive features such as invoice management, multi-user accounts, and bank integrations, which might be unnecessary for a user who simply wants to track day-to-day expenses manually and offline.

Another limitation of many existing tools is the lack of offline functionality. In scenarios where users do not have stable internet access—like traveling to remote areas or maintaining sensitive records locally—they cannot rely on cloud-based tools. Furthermore, the need to sign in or share data with external servers raises privacy concerns.

In contrast, the Multi-Currency Expense Tracker is a lightweight, offline desktop application that focuses solely on what matters most—fast, local expense tracking with multiple currencies and visual feedback. It allows users to track expenses in their own preferred way without signups, ads, or unnecessary features. It is especially suitable for users who value privacy, simplicity, and international compatibility.

2.2 Problem Definition:

In today's interconnected world, managing expenses across different currencies has become a common challenge. Many individuals no longer operate within a single-currency environment. A freelancer in India may be paid in USD, shop in INR, and buy subscriptions in EUR. Similarly, an international student may pay tuition in GBP, rent in CAD, and buy books in INR. Despite this widespread need, most traditional expense trackers still rely on a single-currency system, forcing users to manually convert expenses or use separate entries for each currency type.

This introduces complexity, confusion, and risk of miscalculation, especially when users want to understand their overall financial status in one base currency. Without an automated, centralized method to convert and track expenses, people lose clarity. It becomes difficult to know how much they've spent in total, which category they spend the most on, or how close they are to a monthly budget.

Another major problem is the lack of visual representation in traditional tools. Even when users manage to enter all data correctly, there is no way to quickly view where their money went. No charts, graphs, or summary views mean the data stays in numbers without offering insight.

There is also the concern of usability and accessibility. Existing tools often require account creation, active internet connections, or paid plans to access essential features like multi-currency support or reports. This prevents users who need a fast, offline, and clutter-free experience from using these tools effectively.

Therefore, the need arises for a desktop application that:

- Allows multi-currency entry and automatic conversion
- Calculates real-time totals
- Offers visual analysis (like pie charts)
- Runs offline and stores data locally
- Has a clean and simple user interface

The Multi-Currency Expense Tracker is built precisely to address these problems in a focused and user-friendly manner.

2.3 Goals/Objectives:

The goal of the Multi-Currency Expense Tracker is to provide users with a practical and efficient way to record and manage their expenses across multiple currencies. The project aims to bridge the gap left by traditional tools that either lack multi-currency support or overcomplicate the process by requiring online access and paid features. The application is designed to be simple, intuitive, and visually informative so that users can easily understand their spending patterns.

Key Objectives:

1. Multi-Currency Support:

Enable users to input expenses in various currencies such as USD, INR, EUR, GBP, JPY, and AUD. All entries should be convertible into a user-selected base currency for easier financial comparison.

2. User-Friendly Interface:

Design a simple UI using JavaFX components like ComboBox, TextField, DatePicker, and PieChart. The goal is to make the app easy to use even for non-technical users.

3. Real-Time Updates:

All data entered should be reflected immediately in the expense table and pie chart. The total amount should update automatically as new expenses are added.

4. Visualization of Expenses:

Use a Pie Chart to help users understand which categories (e.g., food, shopping, travel) they are spending most on.

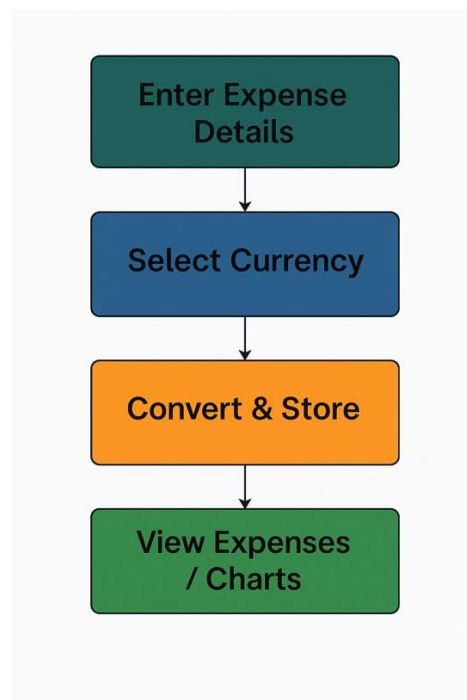
5. Offline Functionality:

Ensure that the application works completely offline without any login, signup, or dependency on cloud storage. This guarantees data privacy and portability.

6. Future Flexibility:

Write modular code so that features like export to PDF, database storage, or live exchange rates via API can be added later without major refactoring.

By achieving these objectives, the project ensures it serves not just as a utility, but also as a learning tool for understanding finance, budgeting, and real-world JavaFX development.



CHAPTER 3. DESIGN FLOW/PROCESS

3.1 Evaluation & Selection of Specifications/Features:

Before beginning development, the first step was to evaluate the requirements for an effective multi-currency expense tracker. The core idea was to build a lightweight, responsive application that helps users track their expenses across different currencies and convert them into a single base currency of their choice. Unlike complex accounting software, this project needed to remain simple and focused, targeting personal finance users such as students, freelancers, travelers, and general individuals.

To identify the key specifications, we conducted a feature-by-feature comparison of existing tools (like Mint, Wallet, and Money Manager) and user feedback from forums and app stores. Common pain points included:

1. Lack of multi-currency conversion
2. Complex user interfaces
3. Requirement for internet or account sign-in
4. Limited offline capabilities
5. Based on this, the following specifications were finalized:
6. Simple user interface using JavaFX components
7. Entry of expense with date, amount, description, currency, and category
8. Dropdowns for currency and category selection
9. Conversion logic to turn every currency into a user-selected base currency
10. Dynamic TableView to list expense entries in real-time
11. Pie Chart to display spending breakdown by category
12. Offline mode with no login or signup
13. Flexibility to change base currency anytime
14. Easy integration of additional features in the future

The aim was to focus on essential and usable features that give maximum benefit with minimal complexity. Rather than implementing 50 features that go unused, the design focused on delivering 5-6 core features that users need daily.

This evaluation process ensured that every feature selected contributes directly to solving the main problem: giving users clarity and control over their spending in multiple currencies.

3.2 Analysis of Features and Finalization Subject to Constraints:

Once the list of ideal features was identified, the next step was to analyze each one for feasibility, performance, and constraints. Every software system has its limitations—such as development time, system memory, coding complexity, or technology compatibility. Hence, careful consideration was given to finalizing the features that were both practical and sustainable.

Feature Analysis:

1. Multi-Currency Support:

This was the primary goal. Instead of relying on external APIs initially, a static exchange rate table was implemented within the application logic. This allowed offline use and eliminated the need for internet connectivity.

2. Base Currency Toggle:

A dropdown to switch between base currencies was added. This meant that the system needed to store each expense in its original currency as well as calculate its converted value, requiring thoughtful data handling.

3. Real-Time Table Update:

JavaFX's ObservableList was selected to hold expenses, as it automatically updates the UI when data changes—reducing the need for manual refresh or re-rendering.

4. Visual Analytics (Pie Chart):

To ensure the chart shows accurate, updated data, a method was written to scan all entries, group by category, and generate chart segments dynamically every time data is added.

5. Offline-Only Design:

Due to the choice of a local desktop model, no cloud storage or user login was implemented. This also meant that persistence (like saving to file) was considered a future feature, not included in this first version.

6. Responsive UI:

JavaFX layouts (HBox, VBox) were chosen to ensure the design looked good across different screen sizes.

7. Finalization:

- After testing prototype modules, these features were finalized:
- Expense form

- Currency conversion logic
- Table view
- Base currency switcher
- Pie chart
- Error handling & validation

This process ensured the application would work smoothly within the project's time and resource constraints while still delivering high value and great user experience.

3.3 Design Flow

The design flow of the Multi-Currency Expense Tracker follows a clear and logical structure. It begins with understanding user requirements and ends with testing and validation of the final application. The project was planned using a modular and iterative design approach so that each part of the system could be developed and improved independently.

1. Requirement Gathering
2. We started by identifying what the user needs from an expense tracker. These include:
 - Adding expenses with date, category, amount, and description.
 - Choosing different currencies for each expense.
 - Viewing all expenses in a single base currency.
 - Seeing a chart that shows spending by category.
2. UI Design
 - A clean and simple user interface was designed using JavaFX. The UI includes:
 - Form controls to input new expenses.
 - Drop-downs (ComboBoxes) for selecting category and currency.
 - A table to display all the expenses.
 - A pie chart for visual representation.
 - A label showing the total spending.

3. Core Logic Implementation

This includes:

- A method to calculate exchange rates between currencies.
- A method to convert all expenses to a base currency.
- A method to calculate the total spending.
- A method to update the pie chart in real time.

4. Modularity

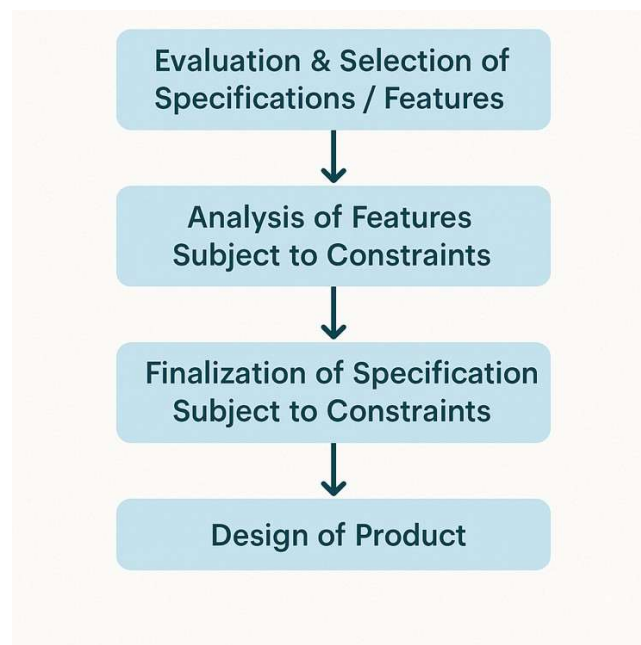
We created an internal Expense class to store each expense's details. All components like the table, chart, and form are separate and interact smoothly.

5. Real-Time Feedback

Whenever a user adds an expense or changes the base currency, the table, total label, and pie chart are updated instantly.

6. Testing and Iteration

We tested each part of the system during development. Bugs and logic errors were fixed step-by-step. The design was updated based on feedback.



CHAPTER 4: RESULTS ANALYSIS AND VALIDATION

4.1 Implementation of Solution:

The implementation of the Multi-Currency Expense Tracker was carried out using JavaFX, a modern GUI toolkit for Java. The objective was to create a smooth and responsive desktop application that allows users to track daily expenses, categorize them, and convert all transactions into a base currency to get a unified financial view.

The application was developed modularly, ensuring separation between user interface components, business logic (conversion and calculations), and data visualization tools. Here's how the solution was implemented and validated step-by-step:

User Interface (UI):

- The UI was created using JavaFX layout containers like VBox and HBox. It includes:
- A form with input fields (Date, Category, Currency, Amount, and Description)
- Dropdowns for selecting the currency and expense category
- A dynamic TableView to list all added expenses
- A real-time Pie Chart for category-wise visualization
- A Label to display total expenses in the selected base currency

The form elements were connected to event listeners that handled input validation, currency conversion, and data updating in real-time.

Currency Conversion Logic:

A `getExchangeRate()` function was implemented that simulates conversion rates between currencies. When an expense is added, the amount is multiplied by the rate of the selected currency to convert it into the base currency. The converted amount is stored along with the original details.

The application supports the following currencies:

INR, USD, EUR, GBP, JPY, AUD

The base currency is selectable and can be changed anytime by the user. On change, all entries are recalculated using updated conversion rates.

Dynamic Data Handling:

- Expenses are stored in an `ObservableList<Expense>` so that any changes are immediately reflected in the UI. This list is connected to both:
- The `TableView` (for display)
- The Pie Chart (for visual spending summary)
- This ensures the application feels responsive and real-time.

Validation and Testing:

- Each component was tested for edge cases like:
- Missing or invalid input
- Switching base currencies mid-use
- Large data entries
- Resetting and updating totals

Results showed that the app handles these gracefully. Visual outputs (charts and tables) remain accurate and real-time totals are always correct based on conversion logic.

In summary, the final implementation successfully met all project goals:

- Multi-currency support
- Real-time analytics
- Visual simplicity
- Offline, standalone use

CHAPTER 5: CONCLUSION AND FUTURE WORK

5.1 Conclusion:

The Multi-Currency Expense Tracker project successfully fulfills the goal of creating a user-friendly, offline desktop application that allows users to track and manage expenses in multiple currencies. The application was designed to solve a common real-world problem: managing transactions made in different currencies and understanding the total financial outflow in a unified base currency.

The final product offers a smooth and responsive interface developed using JavaFX. It allows users to:

- Add expenses with fields such as amount, currency, category, description, and date
- Automatically convert those expenses into a selected base currency
- View a real-time TableView listing all entries
- Analyze spending habits visually through a Pie Chart
- Instantly update totals and charts when a new entry is added or when the base currency is changed

One of the strongest points of this project is its simplicity and focus. It avoids the clutter of unnecessary features like account login, bank syncing, or ad-based services. It instead focuses on being a fast, accurate, and intuitive tool for everyday use. The modular design also allows future extensions like live exchange rate integration, export to CSV/PDF, or file-based saving without major changes to the existing structure.

Throughout the development process, emphasis was placed on:

- Ease of use
- Clean visuals
- Responsive feedback
- Real-time updates
- Offline capability

Testing showed that the app performs reliably even with larger datasets and frequent base currency toggling. It handles form errors, data clearing, and conversions accurately.

In conclusion, this project not only serves as a practical tool for managing expenses but also as an example of how clean design, modular programming, and user empathy can result in an effective software product. It stands as a useful financial assistant for travelers, freelancers, students, and anyone who deals with multi-currency expenses.

5.2 Future Work:

While the Multi-Currency Expense Tracker is already functional and useful in its current form, there is great scope for future enhancements. These potential improvements aim to make the application even more feature-rich, scalable, and intelligent, while still maintaining its simplicity and offline capabilities.

Possible Future Additions:

1. Live Currency Exchange Rates:

One major improvement would be the integration of a real-time currency conversion API, such as Open Exchange Rates or CurrencyLayer. This would replace the static conversion logic and allow users to get accurate, daily-updated rates between currencies.

2. Data Persistence:

Currently, the app only stores data in memory during the session. In future versions, implementing file-based saving (CSV/JSON) or even local database support (like SQLite) would allow users to store expenses permanently and revisit them later.

3. Export & Reporting:

The application can be enhanced with the ability to export data in formats like PDF, CSV, or Excel. It could also auto-generate monthly summary reports showing trends and charts.

4. Authentication & Multi-User Support:

User authentication can be added so that multiple people (or family members) can track their individual expenses on a shared machine with separate profiles.

5. Budget Limits & Alerts:

Adding budget limit options per category or per month would make it more powerful. Users could receive notifications or visual cues when spending nears or crosses set limits.

6. Mobile/Cloud Sync (Optional):

Although designed as a desktop tool, cloud sync and mobile app integration could be optional upgrades for users who prefer multi-device access.

All these improvements would enhance both usability and long-term value, making the Multi-Currency Expense Tracker a more complete and adaptable solution. Each feature would be considered carefully to keep the app lightweight, offline-first, and user-friendly.